

Traceable Reproduction Audit and Feedback-Assisted Semantic-MPC for VLMPC-Style Language-Table Pushing

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Abstract—Vision-language model predictive control (VLMPC) links semantic perception with receding-horizon control, but a reproduction study must distinguish an executable implementation from reliable closed-loop behavior. This paper audits a VLMPC-style Language-Table pushing system under a traceable evaluation protocol covering language and image inputs, parameter ablations, event-triggered VLM re-query, and target-image grounding. The closest unmodified video-prediction MPC reproduction executes and logs raw planner actions, but reaches 0/4 controller-comparison and 0/8 parameter-ablation successes at the $d_{\text{world}}^{\text{final}} \leq 0.08$ evaluation tolerance. We then evaluate feedback-assisted diagnostic controllers that use privileged simulator-state feedback. Grounded original MPC obtains 4/4 and 8/8 diagnostic execution successes in the corresponding controller and ablation suites; all 12 runs replace 23/23 raw planner actions at execution. The latest semantic-MPC suite obtains 18/18 execution successes with mean final distance 0.0641 ± 0.0150 (sample SD) and maximum 0.0798. This success is threshold-dependent: at a stricter 0.05 threshold, semantic MPC succeeds in only 3/18 runs. A new target-image end-to-end matrix covers three target-image cases and three seeds (9 runs): VLM selection is correct in 8/9 runs, and the joint semantic-plus-control success is 7/8 among active trials after separating one pre-satisfied run. A paired semantic-fault audit recovers the correct target in 3/3 forced re-query trials and reaches the true-target threshold in 2/3, versus 0/3 without re-query; this is a controlled repair audit, not evidence of natural-event causal benefit. Fresh matched event controls still show no measurable recovery advantage. The resulting contribution is a negative unmodified reproduction audit plus a traceable feedback-assisted diagnostic stabilization study, not a benchmark-level proof of unmodified VLMPC success.

Keywords—Vision-language model predictive control, semantic MPC, Language-Table, reproduction audit, event re-query, target-image grounding.

I. INTRODUCTION

Robotic manipulation systems increasingly need to connect semantic understanding, visual perception, and closed-loop control. A table-top pushing task appears simple, yet a successful controller must recognize the relevant object, ground the goal, choose a feasible push direction, recover from perception drift, and stop near the target region. Model predictive control (MPC) is useful in such settings because it repeatedly predicts future states, optimizes candidate controls, and executes only

the first action before replanning [1], [2]. Classical MPC, however, usually assumes that state variables and objectives are already available. Vision-language models (VLMs) and vision-language-action systems add semantic grounding and instruction understanding [3]–[5], but they still require reliable low-level control to produce stable physical behavior.

VLMPC aims to bridge these two sides by placing VLM-based semantic evaluation inside a visual MPC loop [6]. The original framework takes a goal image or language instruction, samples candidate action sequences, predicts future visual observations, and evaluates the predictions with a hierarchical cost. Later trajectory-level variants further suggest that reducing the burden of video prediction can improve long-horizon manipulation planning [7]. For a constrained local reproduction study, the key question is therefore not only whether the program can run, but what experimental evidence supports each success claim.

This paper summarizes our study around that question. We use Language-Table push-to-corner tasks and evaluate four components: the unmodified video-prediction MPC chain, a grounded original MPC diagnostic controller, a semantic MPC diagnostic controller, and an event-triggered VLM re-query mechanism. The central reporting choice is to keep negative results visible. The unmodified reproduction is executable and logged, but it remains unsuccessful under the archived threshold. The successful results come from feedback-assisted diagnostic paths and are therefore reported with a privileged-state flag. In particular, the 18/18 and 4/4 positive numbers are execution or semantic-selection results for diagnostic branches, not successful reproductions of the original VLMPC controller. This separation is important because otherwise a stabilized engineering path could be mistaken for a faithful reproduction of the original video-prediction controller.

The contributions are:

- 1) A traceable reproduction audit of a VLMPC-style Language-Table manipulation chain, including unmodified controller comparison and original parameter ablation.
- 2) A conservative symptom analysis that connects failed runs to logged tracking jumps, visual-prediction/control

mismatch, parameter sensitivity, and absence of final-action feedback, without claiming single-factor causality.

- 3) Two bounded diagnostic enhancements: grounded original MPC, which preserves raw planner calls but applies final action correction, and semantic MPC, which combines semantic target selection with state-feedback pushing.
- 4) A branch-level accounting of planner authority, including the grounded branch’s raw-action replacement rate and the semantic branch’s zero active raw-sampling budget.
- 5) A file-traceable experimental package with summary CSVs, metrics JSON files, logs, figures, qualitative videos, matched event controls, a repeated target-image end-to-end matrix, and a controlled semantic-fault re-query audit.

II. RELATED WORK

A. Visual MPC and Video-Prediction Control

Visual MPC uses learned visual dynamics to support model-based manipulation directly in image space. Deep visual foresight and related visual MPC systems showed that action-conditioned video prediction can guide robot motion without a hand-designed dynamics model [8], [9]. Such approaches are attractive for open-ended manipulation, but they are also sensitive to prediction bias, object occlusion, and long-horizon compounding errors. These limitations are relevant to our reproduction: the planner can generate candidate futures and rank actions, but the unmodified closed loop does not reliably drive the object to the selected goal region.

B. Vision-Language Manipulation

Language-conditioned manipulation has used VLMs and language models for task grounding, affordance selection, reasoning, and failure interpretation. CLIPort separates semantic “what” reasoning from spatial “where” execution [10]. SayCan grounds language plans with affordances [3]. Inner Monologue emphasizes feedback-aware embodied reasoning [5], and VoxPoser composes 3D value maps for robotic manipulation from language-model reasoning [11]. These systems motivate our use of explicit feedback and event handling: semantic decisions must be connected to low-level execution evidence, not only to high-level task labels.

C. VLMPC, Trajectory Variants, and Testing

VLMPC integrates VLM perception with MPC-style candidate action evaluation [6]. Traj-VLMPC further reduces computational overhead by using trajectory generation rather than full video prediction [7]. At the same time, evaluation work such as VLATest argues that VLM/VLA manipulation systems require robustness and reliability tests beyond isolated demonstrations [12]. Failure-reasoning systems such as AHA also show the value of identifying manipulation failure modes [13]. Our work follows this evaluation-oriented line: we do not hide failed unmodified runs, and we mark diagnostic controllers that use privileged state feedback.

III. EVALUATION SETUP AND EVIDENCE SOURCES

A. Task Definition

The experiment uses Language-Table desktop pushing tasks. A target object such as *red moon*, *blue cube*, or *yellow pentagon* should be pushed toward a designated corner. The controller input can be a fixed target label, a natural-language instruction, a target image, or a VLM-based scene selection request. The action is a planar end-effector push in the simulated table environment.

The primary metric is the final world-coordinate distance between the target object and the goal:

$$\text{success} = \mathbb{I} [d_{\text{world}}^{\text{final}} \leq 0.08]. \quad (1)$$

The threshold 0.08 is an evaluation tolerance used consistently in the archived final runs, not a universal manipulation success criterion. Threshold sensitivity changes the interpretation of the positive results: at 0.05, the archived grounded diagnostic suites drop to 0/4 and 0/8, and the latest semantic MPC suite succeeds in only 3/18 runs. We therefore report success rates primarily as evidence of coarse task completion under the evaluation tolerance, not precision placement.

For the new target-image matrix, we distinguish a *pre-satisfied* run, for which the true target is already within the threshold before the first action, from an *active control* trial, which starts outside the threshold and executes at least one action. *Joint active success* requires both correct target-image semantic selection and active control success. This prevents a zero-step terminal state from being counted as controller evidence.

B. Evidence Artifacts

All numerical results in this paper are extracted from the study package. The main evidence files are:

- `summary.csv` files for unmodified controller comparison, unmodified parameter ablation, grounded original completion, and semantic MPC full rerun;
- `metrics.json` files containing final distance, runtime, action counts, VLM requests, re-query counts, and raw planner/override counts;
- `log.txt` files containing VLM responses, object-position jumps, and event-trigger messages;
- supplement CSVs for matched event controls, the nine-run target-image matrix, the semantic-fault re-query audit, exact method parameters, checkpoint/code hashes, and VLM cache status;
- GIF/MP4 outputs and key frames used as qualitative evidence;
- robustness-edge audit CSVs covering occlusion, view-point proxy, brightness shift, and unknown-target rejection.

The reporting goal is to make every claim traceable to these artifacts rather than to a post-hoc manual description. The final online VLM path is logged as an OpenAI-compatible backend using the responses wire API and a `gpt-5.5`-labeled gateway model alias. We report this alias because it affects

reproducibility, while excluding the API key and gateway base URL from the paper text. The revision controls use strict no-cache mode and record zero cache hits. They still do not prove that natural re-query causes recovery because matched no-event controls also succeed. The public revision artifact is available at <https://github.com/yixiaogithub66/vlmpc-language-table-reproduction-audit> (release v1.0.4). It contains the authoritative result files, a reproduction guide, parameter records, publication-figure source and exports, separate runtime-asset and source-snapshot manifests, and a reviewer-facing source snapshot. Third-party model weights are not redistributed and must be obtained from their original sources.

IV. METHOD

A. Unmodified Reproduction Chain

The unmodified reproduction follows a VLMPC-style loop. A task input is grounded to a target, candidate action sequences are sampled, a visual prediction model forecasts future observations, and a cost function ranks the candidates before the first action is executed. Controller comparison includes baseline, improved, literature, and literature-v2 settings. Parameter ablation varies sampling budget and cost settings.

In this branch, no final-action state-feedback correction is applied. This distinction is crucial: these runs represent the closest archived reproduction of the original video-prediction MPC chain in our project, and they are evaluated separately from later enhancements.

B. Archived Implementation Parameters

Table I summarizes the recoverable implementation parameters. The unmodified planner uses horizon $H = 20$, action dimension 2, and a correlated-noise sampler with $\beta = 0.5$. The command-line defaults are 200 samples and replan frequency 5, while the archived ablation set contains eight logged configurations rather than a complete 3×3 grid. The literature-style cost is

$$J = d_{\text{obj},\text{goal}} + 0.35d_{\text{contact}} + 0.15d_{\text{arm,obj}} + 18.0J_{\text{align}} + 24.0J_{\text{obs}} - 1.2\Delta d + 12.0\mathbb{I}[\Delta d < 1.0], \quad (2)$$

with obstacle decay 42 pixels. In the close-range branch, the contact, alignment, progress-reward, and non-progress terms are changed to 0.10, 6.0, 1.8, and 20.0, respectively. These values are implementation parameters recovered from `vlmpc.py`, not tuned benchmark claims.

The runtime-asset manifest records SHA-256 prefixes for the externally obtained video-prediction, detector, and tracker assets. The source manifest separately records the executed-source and public-snapshot hashes. Five listed Python files are byte-identical. For `main.py`, the executed-source prefix is `d8078e7cba86` and the public-snapshot prefix is `f98caece7442`; the difference is limited to replacing a drive-local dependency fallback with environment-based configuration, without changing controller logic.

TABLE I
RECOVERABLE IMPLEMENTATION PARAMETERS FROM CODE, METRICS,
AND SUPPLEMENT CSVs.

Parameter	Archived value or rule
Environment	BLOCK_8 Language-Table, 10 Hz, seed 42; archived semantic audit uses 42/43/44; revision matrix uses 45/46/47
Raw MPC	action dim. 2, horizon 20; sampled candidates clipped to $[-0.02, 0.02]$, executed action/feedback bound $[-0.08, 0.08]$
Budgets	default 200/5 samples/plan frequency; eight archived ablations: pf1/ns5, pf2/ns5, pf2/ns10, pf3/ns10, pf2/ns15, cost history low, cost default, cost target high
Detector/tracker	YOLO conf. 0.5; Pysot SiamRPN tracker
Cost weights	contact/approach/alignment/obstacle = 0.35/0.15/18.0/24.0
Progress terms	reward 1.2, non-progress penalty 12.0, min gain 1.0, obstacle decay 42 px
Close range	distance 135 px; contact/alignment/reward/penalty = 0.10/6.0/1.8/20.0
Feedback push	contact offset 0.06, clearance 0.12, move step 0.06, push step 0.075, tolerance 0.018
Original event	default window/min-progress/min-step = 3/1.0/3; matched controls use 2/25.0/3
Semantic event	window 4, min progress 0.002, distance floor $1.5 \times$ success distance
Online VLM	OpenAI-compatible responses API with gpt-5.5-labeled gateway model; API key and base URL excluded
Success threshold	evaluation tolerance $d_{\text{world}} \leq 0.08$; strict audit also reports 0.05 and 0.10

C. Grounded Original MPC

Grounded original MPC is a diagnostic enhancement. The raw planner path still runs, and raw MPC actions are recorded. After the raw action is obtained, a simulator-state feedback push correction is used for the executed action. The correction is bounded by action clipping and uses object, end-effector, and goal positions available from the simulator state. Across the 12 grounded runs (4 controller-comparison runs and 8 ablation runs), the logs contain 23 raw planner actions and 23 feedback overrides per run: the executed-action replacement rate is therefore 23/23, or 100%, in every run. The raw planner remains useful for traceability and branch comparison, but it is not the authority over the executed action in this diagnostic branch.

The grounded and semantic branches use simulator-state feedback and therefore answer a diagnostic question rather than a reproduction question. The feedback action uses object, end-effector, and goal positions from simulator state. Let $\hat{d}_{\text{push}} = (x_{\text{goal}} - x_{\text{obj}}) / \|x_{\text{goal}} - x_{\text{obj}}\|$ and $x_{\text{contact}} = x_{\text{obj}} - 0.06\hat{d}_{\text{push}}$. The controller first stages the end effector near a side-clearance point, then moves to contact, and finally pushes:

$$u_t = \text{clip}(\alpha\hat{d}_{\text{push}} + \beta(x_{\text{contact}} - x_{\text{ee}}), -0.08, 0.08). \quad (3)$$

In the push phase, $\alpha = 0.075$ and $\beta = 0.45$; the stage/contact phases instead use a step-limited move of at most 0.06.

D. Semantic MPC

Semantic MPC is a second diagnostic path. It uses semantic target selection from fixed labels, language instructions, target images, or VLM scene selection, and then performs low-level pushing with state feedback. The strict repair adds an `auto|vlm|vision target-image selector`: in `auto` mode

the system sends the target image and the current scene image to the configured VLM and asks it to choose among visible scene candidates. The outer loop retains repeated receding-horizon execution and logging, but the final semantic push uses no active raw video-prediction sampling budget ($N_{\text{raw}} = 0$); the state-feedback law, rather than a cost-ranked raw action, determines the executed push. It is therefore a stabilized semantic-control baseline within this project, not a faithful reproduction of the raw video-prediction MPC chain.

E. Event-Triggered VLM Re-query

The event mechanism is designed to avoid repeated VLM calls unless tracking or progress conditions become unreliable. In the matched controls, the natural re-query is not forced: `forced_initial_requery` is false for the no-event and event-enabled rows. The strict 2026-06-20 controls run with cache disabled, record zero cache hits, and log fresh responses through the responses wire API. This supports an implementation claim that the event path can fire in matched settings. Because no-event and event-enabled controls both succeed with identical final distances for each seed, the data do not support a causal recovery claim.

To make the recovery question measurable without relabeling a naturally successful run, the revision adds a separate paired semantic-fault audit. Each seed starts with the true target replaced by a deliberately wrong initial target (blue cube instead of red moon). The treatment forces one explicitly labeled synthetic audit re-query at step 3; the control keeps the wrong target for the run. Evaluation is performed against the independent true target, and synthetic re-queries are reported separately from natural detector events. This audit tests whether the implementation can repair a known semantic corruption; it does not estimate natural-event sensitivity or establish a general causal benefit.

V. EXPERIMENTS

A. Experiment Groups

We organize experiments into two groups. **Unmodified reproduction** contains the controller comparison and original parameter ablation. These runs use the reproduced video-prediction MPC chain without final-action state feedback. **Feedback-assisted diagnostics** contains grounded controller comparison, grounded ablation, event re-query, and semantic MPC. These runs use privileged state feedback and are evaluated as stabilization diagnostics.

B. Unmodified Reproduction Results

Table III shows the unmodified reproduction results. Both suites fail under the 0.08 threshold. The final distances are not borderline: the mean values are around 0.25, far above the success threshold. The threshold sensitivity also remains 0% at 0.10. This is the main negative result of the project and should be preserved in any final presentation.

C. Feedback-Assisted Diagnostic Results

Table IV reports the feedback-assisted diagnostic controllers. These results are stable under the 0.08 evaluation threshold, but the privileged-state column changes their interpretation. Grounded original MPC reaches 4/4 successes in controller comparison and 8/8 successes in ablation. The identical grounded final distances should be read cautiously: after privileged final-action feedback is enabled, the terminal push correction dominates the last phase, so the tested raw-planner budget and cost variations affect runtime and traces more than terminal distance. The latest strict semantic-MPC suite reaches 18/18 execution successes across multiple input forms, targets, seeds, and semantic parameters, with final distances between 0.0297 and 0.0798. Across the four semantic conditions, the reported sample SDs range from 0.0041 to 0.0233; the seed-only subset is 0.0587 ± 0.0233 for seeds 42/43/44. The threshold sensitivity is a major caveat rather than a secondary detail: at 0.05, the latest semantic-MPC suite is only 3/18 successful. Execution success is therefore reported separately from unmodified VLMPC reproduction success. The new target-image matrix adds three cases and seeds 45/46/47. It obtains 8/9 correct VLM selections and 7/8 joint active successes after excluding one pre-satisfied run; its all-run final distance is 0.0967 ± 0.1010 , dominated by the single wrong-selection failure. The paired semantic-fault audit recovers the correct target in 3/3 treatment runs and reaches the true-target threshold in 2/3, compared with 0/3 without re-query.

D. Per-Run Evidence Digest

Table VI gives a compact digest extracted from the portable CSVs. The full data files keep run IDs, labels, final distance, steps, runtime, VLM request counts, target-selector metadata, and sanitized provenance identifiers. The table highlights two facts. First, every unmodified run remains well outside the 0.08 threshold. Second, the strict repair improves semantic-MPC execution and target-image semantic selection, but the success claim remains bounded by the use of privileged simulator-state feedback.

E. Event Re-query Evidence

The event evidence is reported as an implementation and control supplement. The archived 2026-06-20 run contains nine rows: three matched no-event controls, three matched event-enabled controls, and three forced re-query audits. The new revision controls add a fresh matched set at seeds 45/46/47; the natural event re-query fires in 2/3 event-enabled rows, while no-event and event-enabled controls reach the same final distance for every matched seed. All revision rows use strict no-cache mode and record zero cache hits. These controls verify trigger execution but do not prove that natural re-query caused recovery. The separate semantic-fault audit below is deliberately evaluated against an independent true target and is reported as a controlled repair test.

The treatment reaches the correct semantic target in all three paired runs, but seed 47 ends at 0.0808 and therefore misses

TABLE II

CONTROLLER-BRANCH RESPONSIBILITIES AND ACTION AUTHORITY. “PRIVILEGED” DENOTES DIRECT SIMULATOR-STATE ACCESS DURING EXECUTION.

Branch	Semantic/planning role	Actual MPC role	Executed action authority	State access
Unmodified VLMPC	Ground target; sample action sequences; predict visual futures	Candidate ranking with a video-prediction cost; first raw action is executed	Raw planner action	Visual observations only
Grounded original MPC	Preserve the same raw planner call and log its actions	Raw planner remains in the loop for comparison and traceability	Privileged state-feedback correction replaces the raw action (23/23 in all 12 runs)	Simulator state
Semantic MPC	Select target from label, language, image, or VLM scene query	Receding-horizon execution/logging; no active raw video-prediction sampling ($N_{\text{raw}} = 0$)	Privileged state-feedback push law	Simulator state
Event-control supplement	Monitor stagnation/semantic events and refresh the VLM instruction when triggered	Event logic controls query timing, not the low-level push optimizer	Same semantic-MPC feedback action	Simulator state

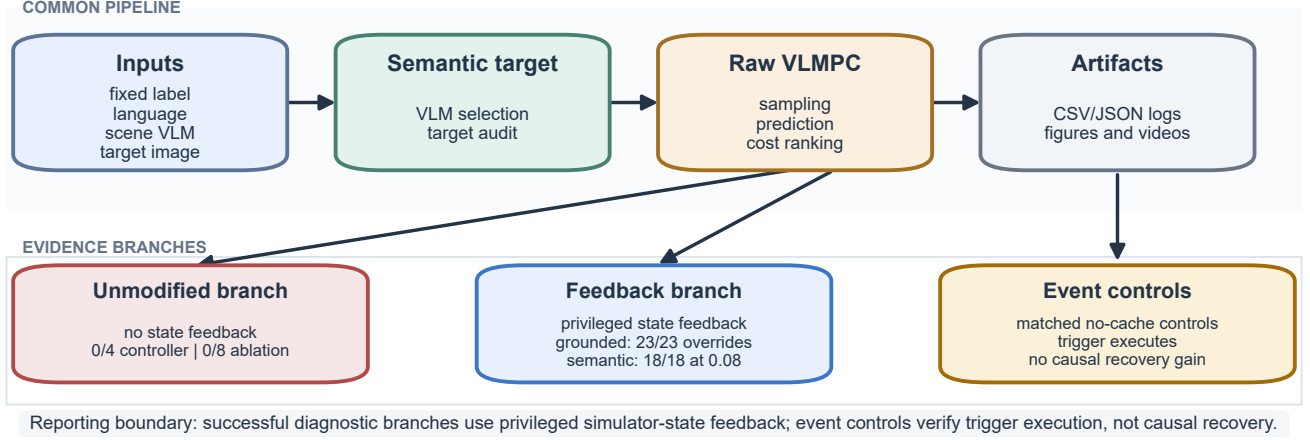


Fig. 1. Strict method overview. The common pipeline is separated from the evidence branches. Grounded original MPC and semantic MPC use privileged state feedback for execution, so their successes are reported as diagnostic stabilization rather than unmodified VLMPC success. The event-control path verifies trigger execution under matched no-cache controls, not causal recovery.

TABLE III

UNMODIFIED REPRODUCTION RESULTS. NO PRIVILEGED STATE FEEDBACK IS USED; FINAL DISTANCE IS MEAN $\pm$ SAMPLE SD.

Suite	Runs	Succ.	Priv.	Final dist.	Max
Controller	4	0/4	no	0.2565 $\pm$ 0.0319	0.2825
Ablation	8	0/8	no	0.2534 $\pm$ 0.0304	0.2939

Threshold sensitivity at 0.05/0.08/0.10 is 0/4, 0/4, 0/4 for controller comparison and 0/8, 0/8, 0/8 for ablation.

the 0.08 threshold by a small margin. The result supports a narrower statement: an explicitly forced re-query can repair a known semantic fault in this setup. It does not establish that the natural stagnation detector finds all such faults, nor that natural re-query improves ordinary successful runs.

F. Target-Image Selection Audit

The target-image interface is now separated into three evidence levels. The old detector/corner heuristic was fragile: it selected *blue cube* for three of the four archived target-image audit cases, giving only 1/4 full-frame heuristic matches. The repaired VLM semantic selector sends both the target image and the current scene image to the configured VLM, asks for a visible scene object, and records the raw response. In the strict 2026-06-20 semantic audit, this selector matches the expected object in 4/4 cases. The archived 18-run semantic-MPC control suite contains one target-image input-form run, which reaches a final distance of 0.0643. The new revision

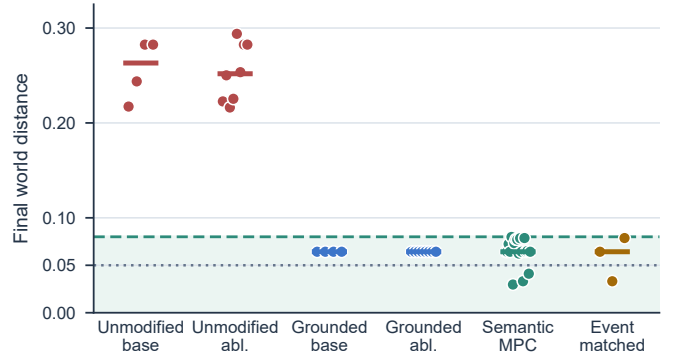


Fig. 2. Final-distance distributions across archived suites. Each dot is one run and the horizontal segment is the within-suite median. The unmodified runs remain far above the 0.08 threshold, while feedback-assisted diagnostics cluster below it.

matrix adds nine end-to-end target-image runs across three cases and three seeds, with 8/9 semantic matches and 7/8 joint active successes. The old four-case audit remains semantic-selection evidence, while the new matrix is still a small diagnostic control study rather than a broad target-image robustness benchmark.

TABLE IV
FEEDBACK-ASSISTED DIAGNOSTIC RESULTS. THESE RUNS USE PRIVILEGED SIMULATOR-STATE FEEDBACK FOR EXECUTION AND ARE NOT DIRECT UNMODIFIED VLMPC REPRODUCTIONS.

Suite	Runs	Exec. succ.	Evidence and semantic check	Final dist.	Priv. state
Grounded controller	4	4/4	target fixed; 23/23 override	0.0643±0.0000	yes
Grounded ablation	8	8/8	target fixed; 23/23 override	0.0643±0.0000	yes
Event controls	9	9/9	trigger logged; cache hits 0	0.0636±0.0184	yes
Semantic MPC latest	18	18/18	target-image audit 4/4; one end-to-end run	0.0641±0.0150	yes
Target-image matrix	9	7/8 active	VLM selection 8/9; one pre-satisfied	0.0967±0.1010	yes
Semantic-fault audit	6 paired	2/3 treatment	synthetic repair 3/3; control 0/3	0.0650±0.0182 treatment	yes

TABLE V
CONDITION-WISE VARIABILITY FOR THE LATEST SEMANTIC-MPC SUITE. VALUES ARE MEAN ± SAMPLE SD OF FINAL WORLD DISTANCE; THE 0.05 AND 0.08 COLUMNS USE $d_{\text{world}}^{\text{final}} \leq$ THE LISTED THRESHOLD.

Condition	n	Mean ± SD	≤ 0.05	≤ 0.08	Target/seed scope	Interpretation
Input forms	4	0.0663 ± 0.0041	0/4	4/4	3 forms + target image	Input-form comparison
Multi-target	8	0.0678 ± 0.0167	1/8	8/8	8 target objects	Small target sweep
Multi-seed	3	0.0587 ± 0.0233	1/3	3/3	red moon, seeds 42/43/44	Seed variability
Semantic ablation	3	0.0568 ± 0.0136	1/3	3/3	default/precise/fast	Semantic-control settings
All latest runs	18	0.0641 ± 0.0150	3/18	18/18	union of above	Bounded diagnostic suite

TABLE VI
PER-RUN DIGEST EXTRACTED FROM ARCHIVED AND LATEST CSV FILES. EXEC. S IS DISTANCE-THRESHOLD SUCCESS UNDER $d_{\text{world}}^{\text{final}} \leq 0.08$.

Suite	Final-distance summary	Range	Exec. S	Semantic/VLM evidence
Unmodified controller	min 0.217; median 0.263; max 0.283	0.217–0.283	0/4	no VLM issue; control fails
Unmodified ablation	min 0.216; median 0.252; max 0.294	0.216–0.294	0/8	no VLM issue; control fails
Grounded controller	all 0.064	0.064–0.064	4/4	fixed target
Grounded ablation	all 0.064	0.064–0.064	8/8	fixed target
Event controls	matched seeds; no-event and event distances identical by seed	0.033–0.079	9/9	fresh responses; no causal gain
Semantic MPC latest	min 0.030; median 0.064; max 0.0798	0.030–0.0798	18/18	archived target-image path 1 run
Target-image revision matrix	8 semantic matches; one wrong blue-cube selection	0.044–0.364	7/8 active	9 runs; one pre-satisfied
Semantic-fault revision audit	treatment repairs target in 3/3; control 0/3	0.045–0.385	2/3 treatment	synthetic re-query; natural count 0

Threshold sensitivity

Unmodified controller (n=4)	0/4	0/4	0/4
Unmodified ablation (n=8)	0/8	0/8	0/8
Grounded controller (n=4)	0/4	4/4	4/4
Grounded ablation (n=8)	0/8	8/8	8/8
Semantic MPC latest (n=18)	3/18	18/18	18/18
	≤ 0.05	≤ 0.08	≤ 0.10

Fig. 3. Threshold sensitivity at 0.05, 0.08, and 0.10, shown as exact success counts. The outlined column is the project threshold. Unmodified runs fail at all three thresholds, while the latest semantic MPC suite is 3/18 at 0.05 and 18/18 at 0.08.

G. Revision Target-Image End-to-End Matrix

The revision repeats the target-image path end to end for three manually specified crops and seeds 45/46/47. The VLM receives the target image and current scene image, selects a named scene object, and the selected target is then passed to

TABLE VII
EVENT RE-QUERY CONTROLS FROM THE STRICT NO-CACHE 2026-06-20 SUPPLEMENT.

Case	Natural query	re- Final dist.	Note
No-event controls, 3 seeds	0	0.0332–0.0787	cache hits 0; fresh VLM count 1
Event-enabled controls, 3 seeds	1 each	0.0332–0.0787	cache hits 0; fresh VLM count 2
Forced scene audit, 3 seeds	audit	0.0644–0.0781	cache hits 0; fresh VLM count 4–6

TABLE VIII
CONTROLLED SEMANTIC-FAULT RE-QUERY AUDIT. THE WRONG INITIAL TARGET IS INJECTED DELIBERATELY; SUCCESS IS MEASURED AGAINST THE INDEPENDENT TRUE TARGET.

Condition	n	Semantic repair	True-target success	Final dist.
No re-query control	3	0/3	0/3	0.2950 ± 0.0868
Synthetic re-query	3	3/3	2/3	0.0650 ± 0.0182

the same semantic-MPC feedback controller. The matrix is intentionally small, but it exposes both a semantic-selection failure and a pre-satisfied terminal state. The blue-cube run at seed 47 selects *blue moon* and remains at distance 0.3644; the yellow-pentagon run at seed 47 is already at distance 0.0645 before acting and is therefore not counted as active control

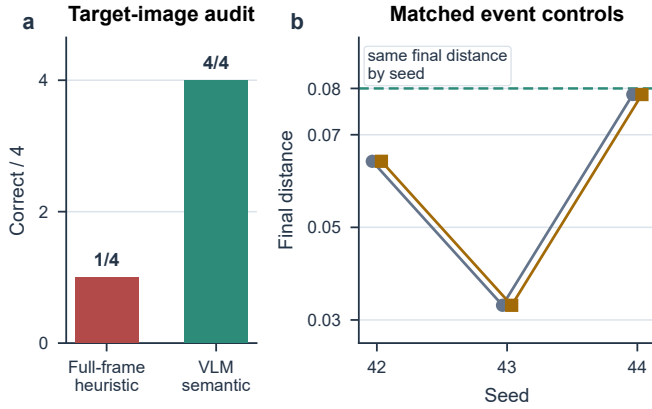


Fig. 4. Target-image and event-control audits. (a) The VLM semantic selector corrects the old full-frame heuristic failure in the four audited target-image cases. (b) Strict no-cache matched controls have identical no-event and event-enabled final distances by seed; all nine archived rows record zero cache hits.

TABLE IX
TARGET-IMAGE SEMANTIC-SELECTION AUDIT. VLM SEMANTIC CHOICE USES THE PROJECT VLM BACKEND WITH TARGET-IMAGE AND SCENE-IMAGE INPUTS.

Case	Exp.	Old	VLM	OK
Red-moon crop	red moon	blue cube	red moon	yes
Blue-cube crop	blue cube	blue cube	blue cube	yes
Yellow-pent. crop	yellow pentagon	blue cube	yellow	yes
Archived frame	red moon	blue cube	red moon	yes

evidence.

The complete per-run table is included in the revision target-image CSV under the sanitized supplement. This expanded matrix strengthens the end-to-end claim relative to the earlier one-run path, while the incorrect blue-cube selection and the pre-satisfied yellow-pentagon run keep the result bounded.

H. Observed Failure Symptoms

The failure analysis is reported as symptom taxonomy rather than causal proof. The unmodified controller comparison and ablation fail together, which indicates that the issue is not solved by the tested parameter variations. The logs also show repeated object-position jumps and tracker fallback. Table XI summarizes the evidence and interpretation.

I. Robustness-Boundary Checks

The robustness-edge audit is not a large benchmark, but it provides boundary evidence for the perception and target-mapping layer. The detector still identifies the red moon under target occlusion, viewpoint proxy, and brightness shift, with confidence decreasing under occlusion. Unknown or unsupported target names are safely rejected rather than mapped to arbitrary known objects.

VI. DISCUSSION

A. Evidence Summary

The study completes a reproducible experimental workflow around the predefined evaluation protocol. The code runs the Language-Table environment, accepts multiple input forms,

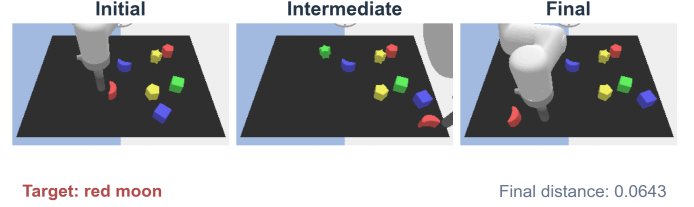


Fig. 5. Representative frames from the latest target-image run. The quantitative claim is taken from CSV/JSON metrics; the frames provide qualitative evidence that the logged run is a real Language-Table execution rather than a table-only report.

logs OpenAI-compatible VLM responses and event triggers, evaluates controller variants and parameter ablations, records summary metrics, and exports figures, GIF/MP4 demonstrations, and final packaging materials. The revision supplement adds fresh no-cache matched event controls, a nine-run target-image end-to-end matrix, a paired semantic-fault re-query audit, checkpoint/code hashes, and exact parameter tables. The analysis preserves the negative unmodified results rather than replacing them with only successful demonstrations.

The strongest positive evidence concerns feedback-assisted stabilization. Grounded original MPC shows that the raw planner can remain in the loop while a final-action feedback correction makes the archived simulator tasks stable under the 0.08 evaluation threshold. The identical grounded-ablation terminal distances are a diagnostic artifact of the shared feedback law, not evidence that planner parameters are irrelevant. The latest semantic-MPC suite achieves 18/18 execution success across several targets, three seeds, four input modes, and semantic parameters. The revised target-image matrix provides 8/9 semantic matches and 7/8 joint active successes, with the failed selection retained rather than removed. The semantic-fault audit shows 3/3 target repair and 2/3 true-target threshold success under an explicitly forced query. These results are useful diagnostic engineering outcomes, but they do not establish benchmark-level reproduction because the successful controllers still use privileged simulator-state feedback.

B. Evidence Boundary

The results do not prove that the original VLMPC paper has been reproduced at its reported performance. Our unmodified archived runs fail. The successful branches use privileged simulator-state feedback, and therefore they are not directly comparable to the original video-prediction MPC results or to published benchmarks. The positive success rates are also tolerance-dependent: they are valid under the 0.08 evaluation threshold but the latest semantic-MPC suite falls to 3/18 at 0.05. The matched natural-event controls show no recovery advantage, while the paired synthetic-fault audit only establishes repair of a deliberately injected semantic error. Finally, target-image grounding should be kept as a small audited capability rather than a broad robustness claim.

TABLE X

REPEATED TARGET-IMAGE END-TO-END MATRIX. FINAL DISTANCE IS MEAN $\pm$ SAMPLE SD OVER ALL RUNS IN THE CASE. ACTIVE TRIALS EXCLUDE RUNS ALREADY WITHIN THE 0.08 THRESHOLD BEFORE THE FIRST ACTION; JOINT ACTIVE SUCCESS REQUIRES CORRECT SEMANTIC SELECTION AND ACTIVE CONTROL SUCCESS.

Target-image case	Runs	VLM semantic match	Pre-satisfied	Active joint success	Final distance	Interpretation
Red moon	3	3/3	0	3/3	0.0693 ± 0.0100	All three active trials succeed
Blue cube	3	2/3	0	2/3	0.1569 ± 0.1799	One wrong selection; failed run remains at 0.3644
Yellow pentagon	3	3/3	1	2/2	0.0641 ± 0.0133	One zero-step pre-satisfied run
All cases	9	8/9	1	7/8	0.0967 ± 0.1010	Small diagnostic matrix, not a benchmark

TABLE XI

OBSERVED SYMPTOMS IN THE FINAL AUDIT. THESE ARE NON-EXCLUSIVE OBSERVATIONS, NOT SINGLE-CAUSE PROOF.

Symptom	Evidence and interpretation
Large unmodified final distance	All 12 unmodified runs end at 0.216–0.294, above 0.08; the reproduced video-prediction MPC chain is executable but not stable in our local setup.
Parameter changes do not rescue the run	Original ablation remains 0/8 across the tested sampling and cost settings; more extensive tuning may still be possible.
Tracking instability	Logs repeatedly report object-position jumps and tracking fallback; this is a plausible contributor, not single-cause proof.
Feedback sensitivity	Grounded diagnostics succeed after final-action state feedback is added; this does not prove raw MPC success.
Semantic mismatch	The old target-image heuristic often selects the wrong object; the repaired selector matches 4/4 archived audit cases and 8/9 revision-matrix cases, with one blue-cube error retained in the report. The evidence is still small and should not be presented as broad robustness proof.

TABLE XII

ROBUSTNESS-EDGE AUDIT FOR TARGET DETECTION AND UNKNOWN-TARGET HANDLING.

Case	Status	Confidence
Original	ok	0.902
Occluded target	ok	0.632
Viewpoint proxy	ok	0.902
Brightness shift	ok	0.900
Purple pole	safe reject	–
Green moon	safe reject	–
Unknown target	safe reject	–

C. Limitations and Future Work

The audit suggests three directions for stronger future work. First, privileged state feedback should be replaced by a visual state estimator or a learned tracker, allowing feedback without simulator oracle access. Our concrete first milestone is to estimate, at each control step, the target centroid (x, y) , end-effector pose, target visibility/confidence, and goal-relative distance from RGB observations; the estimator should be evaluated against simulator state with centroid error (pixels and normalized table coordinates), end-effector position error, goal-distance MAE, visibility F1, and one-step temporal consistency. We would then compare visual-only feedback against oracle feedback using final distance, success at 0.05/0.08, action-replacement rate, recovery latency after an induced tracking jump, and runtime per step. The implementation route is: (i) export synchronized RGB/state pairs from the existing logs, (ii) train or calibrate a detector-plus-tracker with held-out seeds and occlusion/brightness perturbations, (iii) replace the oracle state accessor behind the same feedback-controller

interface, and (iv) run matched visual/oracle trials with fixed seeds and confidence-gated fallback.

Second, event-triggered re-query should be tested over more instructions, perturbations, and no-requery/forced-requery/natural-event settings. The revision’s synthetic-fault audit establishes a measurable repair case but uses a forced query and therefore does not replace a natural-event causal study. A future study must induce a documented tracking or semantic failure, randomize the failure time, and compare enough matched seeds using final distance, recovery success, re-query count, recovery latency, and added VLM cost.

Third, target-image grounding should be evaluated with a larger set of image queries, cropped-object references, full-frame references, and manually verified semantic labels. These additions would convert the current diagnostic report into a stronger experimental paper without changing the interpretation of the current evidence.

D. Data and Code Availability

The public revision artifact at <https://github.com/yixiaogithub66/vlmpc-language-table-reproduction-audit> (release v1.0.4) includes the manuscript source, one authoritative 18-run Semantic MPC file, regenerated threshold and condition statistics, sanitized per-run and aggregate CSVs for the target-image matrix and semantic-fault audit, exact method parameters, a reproducible figure-rendering script, separate runtime-asset and source-snapshot manifests, an automated consistency check, and a reviewer-facing source snapshot. It deliberately excludes API credentials, private gateway addresses, local absolute paths, virtual environments, raw service logs, and third-party model weights. The reported aggregates and table values can be recomputed from the included summaries without relying on cached VLM responses. End-to-end reruns additionally require independently obtained detector, tracker, and video-prediction checkpoints and a separately configured compatible VLM backend, as documented in the reproduction guide.

VII. CONCLUSION

This paper presents a traceable audit and strict repair of a VLMPc-style Language-Table project. The unmodified video-prediction MPC chain is executable but fails in the archived controller comparison and parameter ablation runs. Feedback-assisted diagnostics stabilize the archived simulator tasks under the $d_{\text{world}}^{\text{final}} \leq 0.08$ evaluation tolerance, including a latest semantic-MPC suite with 18/18 execution successes,

a nine-run target-image matrix with 8/9 semantic matches and 7/8 joint active successes, and a paired semantic-fault audit with 2/3 true-target successes after forced re-query. These successes rely on privileged simulator-state feedback and remain tolerance-dependent, with only 3/18 semantic-MPC successes at 0.05. The revision controls do not show a natural-event recovery advantage. The study outcome is therefore best understood as a negative reproduction audit plus a diagnostic repair with explicit limitations, not as a successful unmodified VLMPC reproduction. Future work should replace simulator-state feedback with the specified visual-state estimator, broaden event-requery ablations with induced failures, expand target-image grounding tests, and report success under stricter tolerances.

REFERENCES

- [1] D. Q. Mayne, J. B. Rawlings, C. V. Rao, and P. O. M. Scokaert, "Constrained model predictive control: Stability and optimality," *Automatica*, vol. 36, no. 6, pp. 789–814, 2000.
- [2] J. B. Rawlings, D. Q. Mayne, and M. M. Diehl, *Model Predictive Control: Theory, Computation, and Design*, 2nd ed. Nob Hill Publishing, 2017.
- [3] M. Ahn *et al.*, "Do as i can, not as i say: Grounding language in robotic affordances," in *Proceedings of the Conference on Robot Learning*, 2022.
- [4] A. Brohan *et al.*, "RT-2: Vision-language-action models transfer web knowledge to robotic control," *arXiv preprint arXiv:2307.15818*, 2023.
- [5] W. Huang *et al.*, "Inner monologue: Embodied reasoning through planning with language models," *arXiv preprint arXiv:2207.05608*, 2022.
- [6] W. Zhao, J. Chen, Z. Meng, D. Mao, R. Song, and W. Zhang, "VLMPC: Vision-language model predictive control for robotic manipulation," in *Robotics: Science and Systems*, 2024. [Online]. Available: <https://roboticsconference.org/2024/program/papers/106/>
- [7] J. Chen, W. Zhao, Z. Meng, D. Mao, R. Song, W. Pan, and W. Zhang, "Vision-language model predictive control for manipulation planning and trajectory generation," *arXiv preprint arXiv:2504.05225*, 2025.
- [8] C. Finn and S. Levine, "Deep visual foresight for planning robot motion," *arXiv preprint arXiv:1610.00696*, 2017.
- [9] F. Ebert, C. Finn, S. Dasari, A. Xie, A. X. Lee, and S. Levine, "Visual foresight: Model-based deep reinforcement learning for vision-based robotic control," *arXiv preprint arXiv:1812.00568*, 2018.
- [10] M. Shridhar, L. Manuelli, and D. Fox, "CLIPort: What and where pathways for robotic manipulation," in *Proceedings of the Conference on Robot Learning*, 2021.
- [11] W. Huang, C. Wang, R. Zhang, Y. Li, J. Wu, and L. Fei-Fei, "VoxPoser: Composable 3d value maps for robotic manipulation with language models," in *Proceedings of the Conference on Robot Learning*, 2023.
- [12] Z. Wang, Z. Zhou, J. Song, Y. Huang, Z. Shu, and L. Ma, "Towards testing and evaluating vision-language-action models for robotic manipulation: An empirical study," *arXiv preprint arXiv:2409.12894*, 2024.
- [13] J. Duan, W. Pumacay, N. Kumar, Y. R. Wang, S. Tian, W. Yuan, R. Krishna, D. Fox, A. Mandlekar, and Y. Guo, "AHA: A vision-language-model for detecting and reasoning over failures in robotic manipulation," *arXiv preprint arXiv:2410.00371*, 2024.